Legible Consensus: Topology-Aware Quorum Geometry for Asymmetric Networks

Tony Mason

March 2026

Abstract

Flexible Paxos requires only that Phase 1 and Phase 2 quorums intersect, leaving the shape of those quorums as a design choice. We show that mapping a crumbling-wall quorum construction to a physically tiered network makes the protocol's safety and liveness properties *legible*: an operator can read which tiers retain global consensus capability directly from the wall structure and the current connectivity state, without runtime probing. Using a 10-node Earth/LEO/Moon/Mars topology as a magnifying glass, we demonstrate that a flat Phase 1 family (requiring all tiers) produces 0% global success during conjunction blackout, while a per-tier crumbling-wall family over the same topology produces 100% at 183 ms latency—the only change is quorum geometry. Under the wall construction, three of four tiers retain global liveness during Mars disconnection; only the disconnected tier loses it. Consensus latency at each tier equals the speed-of-light round-trip to the Earth row: 131 ms from LEO, 5.1 s from Moon. A comparison between sparse and full-coverage network topologies separates the wall's quorum obligations from network reachability as independent constraints on liveness. All results are design-level and confirmed across 50-seed sweeps with negligible variance; quorum intersection is verified exhaustively in TLA+. The source repository, sweep data, and reproduction commands are available at `https://github.com/fsgeek/eidolon`.

1 Introduction

Consensus protocols are usually studied over networks where all participants are roughly equidistant. In practice, many systems span topologies with structured asymmetry: datacenters at different distances, edge nodes behind slow links, or—in the extreme case that motivates this paper—tiers separated by minutes to hours of light-travel time.

The naive approach to consensus over such a topology is a flat quorum: majority (or any uniform threshold) over all nodes. This forces the commit path through the slowest tier on every operation. Flexible Paxos [10] showed that safety requires only cross-intersection between Phase 1 and Phase 2 quorum families, not universal majority. This decouples the two phases and opens a design space: Phase 2 (the hot commit path) can be small and fast while Phase 1 (the rare election path) can be large and slow, as long as they intersect.

But how should that design space be explored? Most Flexible Paxos deployments choose quorum sizes—a number for each phase—without considering the *shape* of those quorums. Peleg and Wool [16] showed that quorum geometry can be engineered around combinatorial structure rather than simple counts, with crumbling-wall constructions that achieve high availability through asymmetric quorum families. This paper composes these two ideas: Flexible Paxos's intersection requirement with Peleg and Wool's wall construction, mapped onto a physically tiered network where the rows of the wall correspond to latency tiers.

The result is a protocol whose failure modes are *legible*. In a flat quorum system, when consensus fails, the failure is opaque—you know a quorum didn't form, but the structure doesn't tell you who lost what capability. In the wall construction, the geometry is a map of the failure. Given a connectivity state (which tiers can reach which), the set of tiers retaining global Phase 1 capability is determined by the wall structure alone, without probing.

We demonstrate this using Eidolon, a discrete-event Paxos simulator, over a 10-node topology spanning Earth (5 nodes), LEO (1), Moon (1), and Mars (3)—the 5/1/1/3 topology. The interplanetary setting is not the point; it is the magnifying glass. The latency asymmetry (sub-second Earth, minutes-to-hours Mars) makes properties visible that exist but are harder to measure in terrestrial deployments.

This paper makes four contributions:

1. **The crumbling-wall construction mapped to physical topology.** Per-tier Phase 1 quorum families where each tier reads downward through the wall. Phase 2 is the Earth row. Intersection holds because every Phase 1 contains at least one Earth node. The construction is verified exhaustively in TLA+ over the full 10-node topology (Section 3).

2. **Legible liveness degradation.** Under Mars conjunction blackout, the wall crumbles from the top: only Mars loses global Phase 1 capability. Earth, LEO, and Moon retain it. Consensus latency at each tier equals the speed-of-light round-trip to Earth: 183 ms (Earth), 131 ms (LEO), 5.1 s (Moon). A flat construction over the same topology produces 0% global success during blackout—the only change is quorum geometry (Section 6).

3. **Wall obligations versus network reachability.** A sparse network topology where LEO sees only 3 of 5 Earth ground stations breaks LEO's liveness despite the wall permitting it. The wall determines quorum *obligations*; the network determines *achievability*. These are independent constraints that compose to determine liveness (Section 6.3).

4. **Crash tolerance and the weakest-link migration.** Relaxing Phase 2 from all-of-Earth to k-of-Earth introduces crash tolerance, but the weakest link migrates from the global quorum to the Earth-local quorum. Under two Earth crashes, the topology-aware global construction maintains 98% success while the flat local construction drops below 50% (Section 7).

Each empirical claim is tied to an executable artifact and output file (Appendix A). All results are design-level; deployment would introduce orbital dynamics, antenna scheduling, and variable link quality not modeled here.

2 Background

2.1 Flexible Paxos

Classic Paxos [12] uses majority quorums for both phases: Phase 1 (prepare/promise) establishes a proposer's right to propose, and Phase 2 (accept/accepted) commits a value. Safety requires that any two quorums intersect, which majority trivially guarantees via the pigeonhole principle.

Flexible Paxos [10] observed that the intersection requirement is *cross-phase*, not within-phase: any Phase 1 quorum must intersect any Phase 2 quorum, but Phase 1 quorums need not intersect each other, nor Phase 2 quorums each other. Formally, safety holds if $q_1 + q_2 > |N|$, where q_1 and q_2 are the quorum sizes for Phase 1 and Phase 2 respectively. This means Phase 2 (the hot path, executed on every commit) can be small if Phase 1 (the cold path, executed only during leader election) is correspondingly large.

2.2 Crumbling-Wall Quorum Systems

Peleg and Wool [16] introduced crumbling walls as quorum constructions that achieve high availability through asymmetric structure. Nodes are arranged in rows of potentially different widths. A quorum is formed by selecting elements from successive rows, reading from top to bottom, with each row's width determining how many elements must be chosen. The "crumbling" refers to rows that may lose elements to failure, narrowing the paths through the wall.

The key property: any two paths through the wall must share at least one node, guaranteeing the intersection that consensus requires.

2.3 Composing the Two Ideas

Grid and crumbling-wall constructions [3, 16] show that quorum geometry can be structured around combinatorial properties rather than simple counts. Flexible Paxos shows that the two phases of consensus can use different quorum families. Our construction composes these: the rows of the crumbling wall correspond to physical tiers of the network, and each tier's Phase 1 quorum family reads downward through the wall from that tier's position. Phase 2 uses only the bottom row (the fastest tier). The intersection guarantee follows from the wall structure: every path through the wall reaches the bottom row, and Phase 2 *is* the bottom row.

3 Construction

3.1 System Model

Let the acceptor universe be

$$N = E \cup L \cup U \cup M,$$

where the tiers are pairwise disjoint and represent Earth, LEO, Moon, and Mars respectively. In the evaluated topology,

$$|E| = 5, \quad |L| = 1, \quad |U| = 1, \quad |M| = 3, \quad |N| = 10.$$

We call this the 5/1/1/3 topology. Tiers are ordered by latency: Earth is the fastest (bottom of the wall), Mars is the slowest (top). The network is asynchronous with configurable per-link delay and jitter. Acceptors are crash-stop and non-Byzantine. Links may be removed during scheduled blackout windows.

The simulator uses three logical consensus scopes:

- **Earth-local**: Flexible Paxos over the five Earth nodes ($q_1 = 4$, $q_2 = 2$).
- **Mars-local**: majority quorum over the three Mars nodes ($q = 2$).
- **Global**: the crumbling-wall construction defined below, over all ten nodes.

3.2 Per-Tier Phase 1 Families

The wall maps tiers to rows, ordered top (Mars, slow) to bottom (Earth, fast). A proposer at tier i forms its Phase 1 quorum by reading *downward* from row i: it needs at least one respondent from its own tier and every tier below.

Initiating tier	Phase 1 scope	Tiers needed	Min size
Mars (row 0, top)	Mars + Moon + LEO + Earth	4	4
Moon (row 1)	Moon + LEO + Earth	3	3
LEO (row 2)	LEO + Earth	2	2
Earth (row 3, bottom)	Earth only	1	1

Let $\mathcal{Q}_1^{(i)}$ denote the Phase 1 family for a proposer at tier i:

$$\mathcal{Q}_1^{(i)} = \{Q \subseteq N \mid \forall j \geq i : \ Q \cap T_j \neq \emptyset\},$$

where $T_0 = M$, $T_1 = U$, $T_2 = L$, $T_3 = E$ and tier indices increase from top (slow) to bottom (fast). For relaxed Phase 2 (Section 7), Phase 1 additionally requires enough Earth nodes to satisfy a pigeonhole intersection constraint.

3.3 Phase 2 Family

The Phase 2 family places the hot commit path entirely on the fast tier:

$$\mathcal{Q}_2 = \{E\}.$$

That is, Phase 2 requires all five Earth nodes (the strict construction). Section 7 relaxes this to k-of-$|E|$.

3.4 Safety: Quorum Intersection

Proposition 1. For every tier i and every $Q_1 \in \mathcal{Q}_1^{(i)}$, $Q_2 \in \mathcal{Q}_2$: $Q_1 \cap Q_2 \neq \emptyset$.

Verification. Every $\mathcal{Q}_1^{(i)}$ requires $Q_1 \cap E \neq \emptyset$ (since Earth is at or below every tier). $Q_2 = E$. Therefore $Q_1 \cap Q_2 \supseteq Q_1 \cap E \neq \emptyset$. □

The intersection follows from the wall's structure: every path through the wall reaches the bottom row, and Phase 2 is the bottom row. We verified this exhaustively in TLA+ by enumerating all valid Phase 1 quorums for each of the four initiating tiers against all Phase 2 quorums, including relaxed and crash-fault variants. The `ExhaustiveIntersection` specification checks strict Q_2, relaxed Q_2 ($k = 4$ and $k = 3$), and crash-fault constructions across all four tiers' Q_1 families. The `QuorumIntersection` specification explores the same property via state-space search (11,789 states). No intersection failure was found. Single-decree Paxos agreement was model-checked under a structurally equivalent reduced topology (6 nodes, 67M states, no violation) and partially explored at full scale (6.5B states, no violation before termination). Specifications are in `tla/`.

3.5 Liveness: Reading the Wall

The wall determines liveness as follows. A proposer at tier i can complete Phase 1 if and only if it can reach at least one node in every tier from i down to the bottom. Phase 2 succeeds if the proposer can reach enough Earth nodes (all five under strict $\mathcal{Q}_2$).

During Mars conjunction blackout, cross-tier links between Mars and the inner system are removed. The connectivity state is:

- Mars $\leftrightarrow$ {Moon, LEO, Earth}: **partitioned**.
- Moon $\leftrightarrow$ LEO $\leftrightarrow$ Earth: **connected**.

Reading the wall against this connectivity:

- **Mars** (needs Mars + Moon + LEO + Earth): **blocked**. Cannot reach Moon, LEO, or Earth.
- **Moon** (needs Moon + LEO + Earth): **works**. All three tiers reachable.
- **LEO** (needs LEO + Earth): **works**. Both tiers reachable.
- **Earth** (needs Earth): **works**. Trivially satisfied.

The liveness failure is scoped to exactly the disconnected tier. The wall crumbles from the top.

This is a *legibility property*: given the wall structure and any connectivity state C, the set of tiers retaining global Phase 1 capability is determined without runtime probing. For any tier i, global Phase 1 succeeds if and only if every tier from i to the bottom is reachable from i in C. Any quorum system admits, in principle, computing quorum validity against a connectivity graph. The wall makes this computation structurally trivial: the question reduces to "is every tier from mine to the bottom reachable?"—an $O(\text{tiers})$ check that requires no enumeration of quorum subsets. This is what distinguishes legibility from mere computability.

3.6 What "Global" Means

When an Earth-based proposer completes both phases using only Earth nodes, the resulting decision is *global* in the Paxos sense: it is durable in the quorum intersection. Any future proposer at any tier, upon completing Phase 1, will discover this value in the Earth acceptors' state and cannot contradict it. Mars does not participate in the decision, but Mars cannot disagree with it. This is the distinction between participation and safety: global consensus does not require that all tiers voted, only that no future quorum can produce a conflicting result.

4 Related Work

Paxos established the standard safety argument for replicated state machines under asynchronous messaging and crash failures [12]. Flexible Paxos showed that safety requires only Phase 1/Phase 2 cross-intersection, not universal majority [10]. Our construction builds directly on that result; the novelty is not in the intersection theorem but in mapping the quorum shape to physical topology and characterizing the resulting liveness properties per tier.

Grid and crumbling-wall quorum constructions [3, 16] demonstrate that quorum geometry can be engineered around combinatorial structure. Howard et al. note explicitly that Flexible Paxos can use such constructions; we take up that suggestion. Peleg and Wool analyzed crumbling-wall availability under random, independent node failures. Our contribution is not the wall itself but the mapping: rows correspond to physical latency tiers, and the resulting per-tier liveness properties under scheduled, topology-correlated disconnection—properties the abstract framework identifies as possible but does not explore. Li, Chan, and Lesani [13] formalize heterogeneous quorum systems in which each process may use a different quorum family, and give necessary and sufficient conditions for liveness; our per-tier Phase 1 families are a concrete, topology-aligned instantiation of that general idea. Hierarchical quorum structures have a longer history in database replication [5] and coordination services; our wall differs in that the hierarchy reflects physical latency tiers rather than organizational or logical grouping.

Geo-distributed consensus provides the closest practical analog. WPaxos [1] uses Flexible Paxos with object-stealing to keep commits local. EPaxos [15] removes the leader bottleneck for non-conflicting commands. Mencius [14] distributes leader responsibility across WAN sites. MDCC [11] achieves single-round cross-datacenter commits. All four optimize for tens-to-hundreds of milliseconds WAN latency and assume all replicas are usually reachable. Our setting differs in two respects: the asymmetry spans three orders of magnitude (sub-second to 1342 s), and the target phenomenon is scheduled total disconnection rather than heterogeneous latency.

Delay-tolerant networking [2, 7] motivates the operating regime but addresses routing and custody transfer, not consensus. CRDTs have been studied in DTN-like settings [8]. SwarmDAG [18] uses extended virtual synchrony for partition-tolerant robot swarms; Tran et al. [17] measure consensus latency under partitioning. Protocol Labs' IPC [6] uses hierarchical subnets for blockchain scaling. NASA's DSN study [4] evaluates database replication for ground stations. Our work occupies a gap: Paxos safety under topology-shaped quorum geometry and scheduled disconnection, with per-tier liveness characterization.

5 Experimental Design

Eidolon is a discrete-event simulator built on SimPy. Acceptors process Prepare and Accept messages with configurable processing delay. Proposers fan out messages to all acceptors and collect responses with a per-phase timeout. The network model assigns entities to named locations with per-link latency, jitter, and optional loss. Blackout is modeled as deterministic link removal; the repeater model substitutes a degraded alternate path. Event ordering is by simulation timestamp; the simulator is deterministic given a seed.

5.1 Topology

Five Earth ground stations (NA-West, Europe, Asia, SA-East, Africa) with cross-continental latencies of 30–120 ms. One LEO satellite with 20–45 ms links to ground stations. One Moon node at 1.28 s one-way. Three Mars nodes at configurable one-way delay (186–1342 s, representing closest approach to opposition). Mars-local links are 5 ms.

We evaluate two network variants:

- **Sparse**: LEO has links to 3 of 5 Earth stations (NA-West, Europe, Asia). Mars has links to 2 of 5 (NA-West, Europe). This models a single LEO satellite with limited ground coverage and a Mars relay with two tracking stations.
- **Full coverage**: LEO and Mars have links to all 5 Earth stations. This models a LEO constellation with global coverage and a deep-space network with worldwide antenna placement.

5.2 Experiments

Three experiment families probe different aspects of the construction.

Fixed parameters across all experiments: blackout starts at 600 s, simulation ends at 4000 s, reconciliation interval 120 s, global timeout 500 s, max 1 Paxos round per attempt. All seeds are 40–89 (50 runs per point). Results are reported as means with 95% confidence intervals.

6 Results

6.1 Flat versus Crumbling Wall: Geometry Alone Changes the Outcome

To calibrate the effect of quorum geometry, we compare two constructions over the identical topology, blackout model, and parameter sweep. The *flat* construction requires Phase 1 quorums that

Table 1: Experimental design.

Experiment	Purpose	Key parameters
Parameter sweep	Confirm structural invariance of Earth-initiated global consensus across Mars delay and blackout duration	Mars delay $\in \{186, 750, 1342\}$ s; blackout $\in \{300, 900, 1800\}$ s; 50 seeds per point
Per-tier sweep	Measure which tiers retain global liveness; compare sparse vs. full-coverage topologies	Initiating tier $\in$ {Mars, Moon, LEO, Earth}; both topologies; 186 s Mars delay; 900 s blackout; 50 seeds
Crash tolerance	Characterize the tradeoff between Phase 2 relaxation (k-of-$\lvert E\rvert$) and crash tolerance	$k \in \{3, 4, 5\}$; 0–2 Earth crashes; 186 s; 900 s blackout; 50 seeds

Table 2: Effect of quorum geometry: flat vs. crumbling-wall Phase 1 over the same topology and blackout model. Earth-initiated global proposer, hard blackout, 50 seeds per point. All 18 parameter points (3 Mars delays × 3 blackout durations) return identical rates.

Construction	During-blackout success	Post-blackout success	Avg latency
Flat Phase 1 (all tiers required)	0.0% ± 0.0%	varies by setting	varies
Crumbling wall (Earth reads Earth only)	100.0% ± 0.0%	100.0% ± 0.0%	0.183 s

span all four tiers (the prior version of this work). The *crumbling wall* construction uses the per-tier Phase 1 families defined in Section 3, with an Earth-based global proposer.

The flat construction produces 0% during-blackout success at every sweep point—a tautology of its own design, since it requires Mars nodes that are unreachable. The crumbling wall produces 100% at every point, because the Earth-based proposer's Phase 1 needs only Earth nodes. The difference is entirely in quorum geometry. Same simulator, same topology, same blackout, same timeout budget.

This demonstrates that the design space within Flexible Paxos's safety constraint is large. Both constructions satisfy the cross-intersection requirement. One produces a protocol that is dead during disconnection; the other produces one that doesn't notice. Quorum *shape* is a first-order design decision.

6.2 Per-Tier Liveness: The Wall Crumbles from the Top

Table 3 reports the central result. Under full network coverage, each tier runs its own global proposer during Mars conjunction blackout (186 s one-way, 900 s blackout, hard blackout model, 50 seeds).

Three of four tiers retain global consensus capability during hard Mars blackout. Only Mars—the disconnected tier—loses it. The liveness failure crumbles from the top of the wall, exactly as the construction predicts.

The latency column maps directly to physics. Moon's 5.131 s is two Paxos phases at the 1.28 s one-way Moon–Earth light delay ($\approx 4 \times 1.28$ s). Earth's 183 ms reflects cross-continental round-trips within the Earth tier. LEO's 131 ms is *faster* than Earth—a counterintuitive result explained by the network topology: LEO-to-ground links (20–45 ms) are shorter than cross-continental Earth links (50–120 ms). Wall position determines quorum *obligations*; the speed of light determines their *cost*. These are correlated but not identical, and LEO is the proof.

Mars fails during blackout for the structural reason predicted by the wall (it cannot reach inner tiers for Phase 1). Mars also fails *before* blackout, for a different reason: Mars-to-Earth round-trip

Table 3: Per-tier global consensus during Mars conjunction blackout (full-coverage topology, hard blackout, 186 s Mars delay, 900 s blackout, mean ± 95% CI over 50 seeds).

Initiating tier	Wall position	During-blackout	Avg latency	Recovery lag
Mars	top (row 0)	0.0% ± 0.0%	—	—
Moon	row 1	100.0% ± 0.0%	5.131 s ± 0.000	6.7 s ± 0.01
LEO	row 2	100.0% ± 0.0%	0.131 s ± 0.000	61.8 s ± 0.01
Earth	bottom (row 3)	100.0% ± 0.0%	0.183 s ± 0.000	62.6 s ± 0.01

Table 4: Per-tier global consensus under sparse topology (same parameters as Table 3).

Initiating tier	Wall prediction	Sparse result
Mars	blocked	0.0% ± 0.0%
Moon	works	100.0% ± 0.0%
LEO	works	0.0% ± 0.0%
Earth	works	100.0% ± 0.0%

time (≈372 s per phase) exceeds the 500 s per-phase timeout budget for a complete two-phase Paxos round. This is not a blackout finding but a physics finding: Mars-initiated global consensus requires either patient timeouts or delegation to a lower-tier proposer.

Recovery lag differences between tiers (6.7 s for Moon vs. 62.6 s for Earth) are scheduling artifacts of the 120 s reconciliation interval interacting with blackout boundaries, not protocol properties.[1]

6.3 Wall Obligations versus Network Reachability

Table 4 repeats the per-tier experiment under the sparse network topology.

LEO drops from 100% (full coverage) to 0% (sparse). The wall says LEO needs LEO + Earth; that obligation is satisfiable. But strict Phase 2 requires all five Earth nodes, and LEO has links to only three ground stations. The wall determines what a tier *needs*; the network determines what it *can reach*. Liveness requires both.

This separation has practical implications. An operator designing a tiered deployment must check two things: (1) the wall structure, which determines quorum obligations per tier, and (2) the network topology, which determines whether those obligations are achievable. The wall is the blueprint; the network is the building site. The legibility property extends to both: given the wall and the connectivity graph, an operator can determine which tiers have liveness without probing.

Moon succeeds in both topologies because it has links to all five Earth ground stations. Earth succeeds because it is co-located with the ground stations. The sparse topology is a realistic model of a single LEO satellite with limited ground coverage; the full-coverage topology models a constellation. The three missing links are the difference between 0% and 100% for LEO-initiated consensus.

[1] Moon's slightly longer per-round time shifts its reconciliation cadence so the next attempt falls shortly after blackout ends. Earth's falls ∼62 s later. The effect is deterministic and independent of the consensus protocol.

Table 5: Coordinated relaxation under Earth crashes (mean ± 95% CI, 50 seeds). Earth-local Flexible Paxos: "std" is q_1=4, q_2=2; "maj" is q_1=q_2=3.

Global $\mathcal{Q}_2$	Local	Crashes	Earth-local	Global during	Global post	Recovery (s)
strict	std	0	100.0%	100.0%	100.0%	488.9±2.6
strict	std	1	100.0%	—	0.0%	n/a
4-of-5	std	0	100.0%	100.0%	100.0%	489.9±2.5
4-of-5	std	1	100.0%	100.0%	100.0%	489.1±2.1
3-of-5	std	2	49.7%	98.0%±3.9%	100.0%	512.7±2.0
3-of-5	maj	2	100.0%	92.0%±7.6%	100.0%	513.4±2.2

7 Crash Tolerance and Coordinated Relaxation

The strict Phase 2 choice $\mathcal{Q}_2 = \{E\}$ introduces a fragility: a single crashed Earth acceptor blocks all global Phase 2 progress. This section relaxes that choice.

7.1 Relaxed Construction

Let the relaxed Phase 2 family require k of $|E|$ Earth nodes. We evaluate $k = 4$ (tolerates one crash) and $k = 3$ (tolerates two). Phase 1 requires correspondingly more Earth nodes to maintain intersection: the minimum Earth count in any Q_1 is $|E| - k + 1$ (2 for $k = 4$, 3 for $k = 3$). Both constructions were verified exhaustively in TLA+.

7.2 Results: The Weakest Link Migrates

Table 5 reports the crash-tolerance sweep (Earth-initiated, repeater-assisted, 186 s Mars delay, 900 s blackout, 50 seeds).

Strict Phase 2 is completely crash-intolerant. A single Earth crash produces permanent global liveness loss (row 2).

The weakest link migrates. At two crashes with standard Earth-local quorums, the *global* construction (3-of-5, spanning all tiers) maintains 98% during-blackout success—but the *Earth-local* construction drops to 49.7% because it cannot form a Phase 1 quorum of 4 with only 3 remaining Earth nodes (rows 5–6). The topology-aware global quorum is more crash-resilient than the flat local quorum because it can recruit acceptors from other tiers.

Coordinated relaxation restores resilience. Relaxing the Earth-local quorum to majority ($q_1 = q_2 = 3$) recovers 100% local success while maintaining 92% global success (row 6). The tradeoff curve:

Regime	Phase 1 min	Phase 2	Crash tol.	Commit cost
Strict ($k = 5$)	6 / 10	5 / 5	0	2-node Earth
Relaxed ($k = 4$)	7 / 10	4 / 5	1	2-node Earth
Relaxed ($k = 3$) + maj	8 / 10	3 / 5	2	3-node Earth

8 Threats to Validity and Limitations

All results are design-level, not deployment-level.

Earth as single point of failure. The wall concentrates both safety and liveness at the bottom row. Every Phase 1 reads down to Earth; every Phase 2 is on Earth. If the entire Earth tier becomes unreachable—not individual node crashes, but tier-level partition—global consensus fails for all tiers. This is the price of placing the anchor at the bottom of the wall. Relaxed Phase 2 tolerates individual Earth crashes but not Earth-tier partition.

Abstract network model. Blackout is deterministic link removal. Orbital dynamics, antenna scheduling, contention, signal coding, and variable link quality are not modeled. The LEO topology variant demonstrates that network reachability matters; a more realistic model with time-varying LEO coverage (orbital ground tracks) would show liveness fluctuating with satellite position.

Stylized workload. Reconciliation runs on a fixed 120 s interval. This creates scheduling artifacts in recovery lag measurements; recovery lag differences between tiers are workload properties, not protocol properties.

Single topology. We evaluate one topology size (5/1/1/3). The construction generalizes to other tier sizes, but we do not sweep Earth-tier size or Mars replication degree.

Crash-stop only. Byzantine behavior, correlated faults, reconfiguration during blackout, and storage failures are outside scope.

Per-tier crash tolerance. The interaction between wall position and crash tolerance (does a Moon-initiated proposer degrade differently from an Earth-initiated one under Earth crashes?) is not explored and remains future work.

9 Discussion

9.1 The Design Space within Flexible Paxos

The 0% $\rightarrow$ 100% transition between flat and crumbling-wall constructions demonstrates that quorum *shape* is a first-order design decision within the Flexible Paxos framework. Both constructions satisfy the cross-intersection safety requirement. The difference is entirely in how Phase 1 quorum families are structured. Most Flexible Paxos deployments choose quorum *sizes*—a number for each phase—without considering the geometry of those quorums. Recent work continues to expand this design space: Howard and Abraham [9] show that even Fast Paxos's quorum constraints are more conservative than necessary. The results here suggest that for topologically asymmetric networks, the shape matters at least as much as the size.

9.2 Topology-Scoped Consistency

During conjunction blackout, the system is not simply "inconsistent." Each tier maintains strong local consistency via its local quorum. Cross-tier state becomes stale in a structured way: each tier's view of remote state reflects the last successful global reconciliation before disconnection. The wall determines which tiers retain the ability to make globally durable decisions during disconnection. This is a topology-scoped view of consistency in which quorum geometry determines which consistency domains survive disruption, and the network topology determines when they reconcile.

The framework generalizes beyond the interplanetary setting. Any system with tiered latency asymmetry—edge computing, multi-region cloud, satellite-terrestrial hybrids—exhibits similar structure. The wall construction makes the consistency topology explicit rather than implicit, allowing operators to reason about failure modes from the quorum design rather than discovering them at runtime.

10 Conclusion

We showed that mapping a crumbling-wall quorum construction to a physically tiered network produces a consensus protocol whose failure modes are legible from the geometry. The wall crumbles from the top: during disconnection, only the unreachable tier loses global liveness. Three of four tiers in the evaluated topology retain global consensus capability during hard Mars blackout, at latencies determined by the speed of light to Earth.

Two additional findings emerged from the experiments. First, the sparse-versus-full-coverage topology comparison separates quorum obligations from network reachability as independent constraints on liveness, a distinction the theory alone does not make visible. Second, the crash-tolerance analysis reveals that topology-aware global quorums are inherently more resilient than flat local quorums, because they can recruit acceptors from other tiers—a counterintuitive consequence of the wall's asymmetric structure.

The interplanetary setting is the magnifying glass, not the point. The underlying principle—that quorum geometry should mirror the topology it serves—applies wherever latency asymmetry and structured disconnection determine which operations are practical.

A Claim-to-Artifact Traceability

Table 6 maps the paper's main claims to executable artifacts. Paths are relative to the repository root.

Table 6: Claim-to-artifact traceability.

Claim	Evidence	Artifact(s)
Flat construction: 0% during blackout across all 18 sweep points.	Old construction sweep results.	`results/step9/` (prior sweep data).
Crumbling wall: 100% during blackout across all 18 sweep points, 183 ms latency.	All rows show 100% during-blackout.	`results/step9_crumbling/step9_sweep.csv`; `experiments/step9_sweep.py`.
Per-tier liveness: 3 of 4 tiers retain global consensus during blackout.	Per-tier sweep, both topologies.	`results/tier_liveness/tier_sweep_ci.csv`; `experiments/tier_liveness_sweep.py`.
LEO faster than Earth (131 ms vs 183 ms).	Latency column in per-tier results.	Same as above.
Sparse topology: LEO drops to 0%.	Sparse rows in per-tier sweep.	Same as above.
Weakest-link migration and coordinated relaxation.	Crash-tolerance sweep rows.	`results/step10/step10_sweep_ci.csv`; `experiments/step10_sweep.py`.
TLA+ verification of quorum intersection.	Exhaustive enumeration, all tiers × all constructions.	`tla/ExhaustiveIntersection.tla`; `tla/QuorumIntersection.tla`.

Claim	Evidence	Artifact(s)
Paxos agreement model-checked.	67M states (reduced), 6.5B states (full, partial).	`tla/PaxosSmall.tla`; `tla/PaxosFull.tla`.

References

[1] Ailidani Ailijiang, Aleksey Charapko, Murat Demirbas, and Tevfik Kosar. WPaxos: Wide area network flexible consensus. *IEEE Transactions on Parallel and Distributed Systems*, 31(1):211–223, 2020.

[2] Vinton Cerf, Scott Burleigh, Adrian Hooke, Leigh Torgerson, Robert Durst, Keith Scott, Kevin Fall, and Howard Weiss. Delay-tolerant networking architecture. Technical Report RFC 4838, RFC Editor, 2007.

[3] S. Y. Cheung, M. H. Ammar, and M. Ahamad. The grid protocol: A high performance scheme for maintaining replicated data. *IEEE Transactions on Knowledge and Data Engineering*, 4(6):582–592, 1992.

[4] Andrea M. Connell. An analysis of database replication technologies with regard to deep space network application requirements. Technical report, Jet Propulsion Laboratory, California Institute of Technology, 2011.

[5] Khuzaima Daudjee and Kenneth Salem. Lazy database replication with ordering guarantees. *Proceedings of the 20th International Conference on Data Engineering (ICDE)*, pages 424–435, 2004.

[6] Alfonso de la Rocha, Lefteris Kokoris-Kogias, Jorge M. Soares, and Marko Vukolić. Hierarchical consensus: A horizontal scaling framework for blockchains. In *IEEE 42nd International Conference on Distributed Computing Systems Workshops (ICDCSW)*, 2022.

[7] Kevin Fall. A delay-tolerant network architecture for challenged internets. In *Proceedings of the 2003 Conference on Applications, Technologies, Architectures, and Protocols for Computer Communications (SIGCOMM)*, 2003.

[8] Frédéric Guidec, Yves Mahéo, and Camille Noûs. Supporting conflict-free replicated data types in opportunistic networks. *Peer-to-Peer Networking and Applications*, 16:395–419, 2023.

[9] Heidi Howard and Ittai Abraham. Fast flexible paxos. *arXiv preprint arXiv:2008.02671*, 2021.

[10] Heidi Howard, Dahlia Malkhi, and Alexander Spiegelman. Flexible paxos: Quorum intersection revisited. In *20th International Conference on Principles of Distributed Systems (OPODIS 2016)*, 2017.

[11] Tim Kraska, Gene Pang, Michael J. Franklin, Samuel Madden, and Alan Fekete. MDCC: Multi-data center consistency. In *Proceedings of the 8th ACM European Conference on Computer Systems (EuroSys)*, pages 113–126, 2013.

[12] Leslie Lamport. The part-time parliament. *ACM Transactions on Computer Systems*, 16(2):133–169, 1998.

[13] Yimeng Li, Andrew Chan, and Mohsen Lesani. Quorum subsumption for heterogeneous quorum systems. In *37th International Symposium on Distributed Computing (DISC 2023)*, volume 281 of *LIPIcs*, pages 28:1–28:19, 2023.

[14] Yanhua Mao, Flavio P. Junqueira, and Keith Marzullo. Mencius: Building efficient replicated state machines for WANs. In *Proceedings of the 8th USENIX Symposium on Operating Systems Design and Implementation (OSDI)*, pages 369–384, 2008.

[15] Iulian Moraru, David G. Andersen, and Michael Kaminsky. There is more consensus in egalitarian parliaments. In *Proceedings of the 24th ACM Symposium on Operating Systems Principles (SOSP)*, pages 358–372, 2013.

[16] David Peleg and Avishai Wool. The availability of quorum systems. *Information and Computation*, 123(2):210–223, 1995.

[17] Jason A. Tran, Gowri S. Ramachandran, Claudiu B. Danilov, and Bhaskar Krishnamachari. An evaluation of consensus latency in partitioning networks. In *MILCOM 2019 – IEEE Military Communications Conference*, 2019.

[18] Jason A. Tran, Gowri S. Ramachandran, Parth M. Shah, Claudiu B. Danilov, Roberto A. Santiago, and Bhaskar Krishnamachari. SwarmDAG: A partition tolerant distributed ledger protocol for swarm robotics. *Ledger*, 4(S1), 2019.